A Self-Contained Proof of the Characterization of Label-Connected Graphs

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Abstract

In the classical gossiping setting, each participant initially knows a distinct piece of information. Whenever two participants call each other, they exchange everything they currently know. An undirected multigraph is label-connected if its edges can be ordered as calls so that every participant learns all information, using each edge at most once. Equivalently, it admits a temporal labeling that makes it connected by strict temporal paths. We give a self-contained proof of the classical characterization: a multigraph is label-connected if and only if it has at most one vertex, has two spanning trees sharing at most one edge, or admits a cycle of length at least four and two spanning forests, all pairwise edge-disjoint, each forest having four components that meet the cycle. The proof constructs a complete call order from either witness and, conversely, obtains a witness by contracting two-tree blocks and lifting a suitable quotient structure.

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1 Introduction

Distributing information among participants is a basic task in communication networks. In a peer-to-peer exchange, two participants can share both their own information and everything they have learned from earlier exchanges. When the available pairwise exchanges are fixed,

their order determines how far information can spread. We ask whether these exchanges can be scheduled so that every participant receives all information, without introducing additional exchanges.

The classical gossiping problem captures this question in a particularly clean form. Each participant initially knows a distinct piece of information, and whenever two participants call each other, they exchange everything they currently know. If every pair may communicate, $2n - 4$ calls are necessary and sufficient for $n \geq 4$ participants [1, 9]. We instead prescribe the available calls by an undirected multigraph: vertices represent participants, and each edge represents one call between its endpoints. The task is to decide whether these calls can be ordered so that all information reaches every participant. Thus the difficulty lies not in selecting sufficiently many calls, but in coordinating a fixed and possibly highly constrained set of calls.

This question is also a temporal graph realization problem. A temporal graph (G, λ) consists of an undirected multigraph $G = (V, E)$ and a labeling $\lambda\colon E \to \mathbb{N}$ assigning a time to each edge. A *strict temporal path* is a path whose edge labels strictly increase in traversal order. The temporal graph is *temporally connected* if such a path exists for every ordered pair of vertices. Each edge receives exactly one label.[1] Different edges may receive equal labels. However, we can give all edges distinct labels while preserving the order of edges whose labels were already different. Every strict temporal path is preserved, so it suffices to consider a total order of the calls (Section 2). Loops are irrelevant, and parallel edges represent distinct calls. A graph with at most one vertex is a yes-instance. Disconnected graphs are immediate no-instances.

Gossiping	
Input:	An undirected multigraph $G = (V, E)$.
Question:	Is there a labeling $\lambda\colon E \to \mathbb{N}$ such that (G, λ) is temporally connected?

Following the classical literature, we call a multigraph *label-connected* if it is a yes-instance of Gossiping. Göbel, Cerdeira, and Veldman [8] proved that recognizing label-connected graphs is NP-complete. A classical characterization describes the yes-instances using either two spanning trees (a *T-witness*) or a cycle together with two spanning forests (an *M-witness*) [3, 4, 11]. We give a self-contained proof of this characterization using the notation and multigraph conventions fixed in Section 2.

First, we construct a complete call order from either type of witness (Section 3). For the converse, we contract maximal vertex sets supporting two edge-disjoint spanning trees and analyze the resulting quotient (Section 4). The main difficulty is choosing a quotient cycle so that its witness can be lifted to the original graph while keeping the cycle and forests edge-disjoint.

The contraction, exchange, and lifting framework follows Burosch and Laborde [3]; the closed-tree counting argument follows Bumby [2] and Kleitman and Shearer [10]. We give these arguments in full and use information paths in the original graph to ensure that the lifted cycle and forests are edge-disjoint.

[1] We use the temporal-graph terminology of edge labels and the gossiping terminology of calls interchangeably: an edge represents a call, and its label specifies when that call takes place.

2 Preliminaries and the Characterization

We fix our graph and temporal-labeling conventions, define the two spanning witnesses, and state the classical characterization. We also recall the basic matroid notions used in the sufficiency proof.

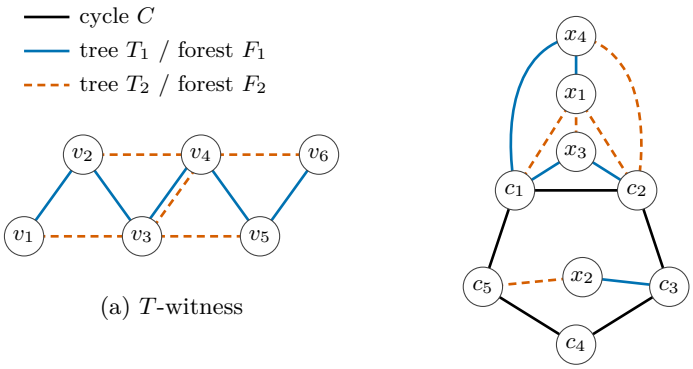
Graphs. We use standard graph-theoretic terminology; see, for example, Diestel [7]. Unless stated otherwise, graphs are finite undirected multigraphs, with parallel edges treated as distinct. We write $n = |V(G)|$ and $m = |E(G)|$. Degrees count multiplicities; the maximum degree of the support is explicitly identified and denoted by Δ. For $X \subseteq V(G)$, let $G[X]$ be the induced subgraph and $E[X]$ its edge set. For $F \subseteq E(G)$, the graph $(V(G), F)$ retains isolated vertices, and $V(F)$ is the set of endpoints of F. We write $[a] = \{1, \dots, a\}$ for a positive integer a, use $G - S$ and $G - F$ for vertex and edge deletion, and $A \setminus B$ for set difference.

The *support* of a multigraph is the simple graph obtained by deleting loops and replacing each nonempty parallel class by one edge. A path has no repeated vertices; a path consisting of one vertex is allowed. A *cycle* of length $\ell \geq 1$ is a closed walk $v_0 e_1 v_1 \cdots e_\ell v_\ell$ with $v_\ell = v_0$, pairwise distinct edges $e_1, \dots, e_\ell$, and pairwise distinct vertices $v_0, \dots, v_{\ell-1}$; we identify it with the subgraph formed by these vertices and edges. Thus a cycle of length one is a loop, and a cycle of length two consists of two parallel edges. The length of a path is its number of edges. A forest contains no cycle; in particular, it contains no loop and no pair of parallel edges. A spanning forest includes every vertex, and isolated vertices count as components. A spanning forest on n vertices with c components has $n - c$ edges. A *matching* is a set of pairwise vertex-disjoint edges. Edge-disjoint subgraphs may share vertices.

We discard loops in instances of Gossiping, since they do not affect communication. Contraction of a connected vertex set identifies its vertices, retains external parallel edges, and deletes the resulting loops. We call the graph obtained by contracting one or more pairwise disjoint connected vertex sets a *quotient* of the original graph. In the graphic matroid constructions, we also identify vertex sets that need not be connected. There we explicitly retain all edges, including resulting loops, to preserve the matroid ground set.

Temporal Connectivity. A general introduction to temporal graphs is given by Casteigts et al. [6]; for the distinctions between simple, strict, and proper temporal graphs, we follow Casteigts, Corsini, and Sarkar [5]. A *temporal labeling* of a multigraph $G = (V, E)$ is a function $\lambda \colon E \to \mathbb{N}$ assigning one positive integer to each edge. A *strict temporal path* traverses edges $e_1, \dots, e_\ell$ with $\lambda(e_1) < \cdots < \lambda(e_\ell)$. The temporal graph (G, λ) is *temporally connected* if a strict temporal path exists from u to v for every ordered pair $u, v \in V$. Paths of length zero are allowed, so a graph with at most one vertex is temporally connected. Order the edges by their labels, placing edges with equal labels in an arbitrary order, and assign distinct labels $1, \dots, m$ in that order. This preserves every strict comparison between original labels and hence every strict temporal path. Thus every temporally connectable graph admits a bijective labeling $\lambda \colon E \to [m]$.

A bijective temporal labeling defines a *gossip scheme*, that is, a total order of the calls. Initially each vertex knows its own distinct secret; when an edge is called, its endpoints exchange all secrets they currently know. After any prefix of the order, a vertex v knows the secret of a vertex u if and only if some strict temporal path from u to v uses only calls of that prefix. A gossip scheme is *complete* if every vertex knows every secret after all calls. Thus (G, λ) is temporally connected if and only if the corresponding gossip scheme is complete.



(a) T-witness

(b) M-witness

Figure 1 The two witness types. (a) A T-witness: two spanning trees T_1 (solid) and T_2 (dashed) that share only the edge v_3v_4. (b) An M-witness on nine vertices: the black edges form the cycle C, and the blue and orange edges form spanning forests with four components each, all meeting C. The graph in (b) has $15 = 2n - 3$ edges, but it is not a T-graph (see the text). We use it as a running example in Figure 2.

Spanning Witnesses. The classical characterization uses the following two types of subgraphs. We follow the terminology of Burosch and Laborde and of Krumme [3, 11].

▶ **Definition 2.1** (T-graph and T-witness)**.** *A multigraph is a* T-graph *if it has two spanning trees sharing at most one edge. We call such a pair a* T-witness*.*

The second witness type replaces the two almost edge-disjoint trees by two forests connected through a common cycle.

▶ **Definition 2.2** (M-witness and M-graph)**.** *An* M-witness *of a multigraph G consists of pairwise edge-disjoint subgraphs C, F_1, F_2, where C is a cycle of length at least four and each F_i is a spanning forest with exactly four components, each containing a vertex of C. A graph is an* M-graph *if it admits an M-witness and is not a T-graph. An M-witness is* minimum *if $|E(C)|$ is minimum among all M-witnesses of G.*

In this terminology, T-graphs and M-graphs form disjoint classes. The gossiping literature calls an M-witness a *realizing set*. We use "witness" to emphasize its algorithmic role. For statements that also apply to T-graphs containing such a cycle-and-forests structure, we say explicitly that the graph *admits an M-witness*.

In intuitive descriptions and figures, we call F_1 the *blue forest* and F_2 the *orange forest*. In figures showing witnesses, we consistently use blue for F_1 and orange for F_2, and the same colors for the corresponding witness trees. We retain the notation F_1, F_2 in formal statements.

Figure 1 shows both witness types. The graph in Figure 1(b) has nine vertices and $15 = 2n - 3$ edges, which is exactly the number of edges of two spanning trees sharing one edge. Nevertheless, it is not a T-graph: the five vertices c_1, c_2, x_1, x_3, x_4 induce nine edges, but at most $2(5 - 1) = 8$ of them lie in a union of two forests. Hence, every union of two forests has at most $2n - 4$ edges. We use this graph as a running example.

Matroids. We use standard matroid terminology; see Oxley [12]. A *matroid* $M = (E, \mathcal{I})$ has a finite ground set E and a family $\mathcal{I} \subseteq 2^E$ of *independent sets* containing $\varnothing$, closed under

subsets, and satisfying the exchange axiom: for $A, B \in \mathcal{I}$ with $|A| < |B|$, some $b \in B \setminus A$ has $A \cup \{b\} \in \mathcal{I}$. We also write $\mathcal{I}(M)$ for this family. A *basis* is an inclusion-maximal independent set; all bases have the same size, the *rank* $r(M)$. A *loop* belongs to no independent set.

The *graphic matroid* $M(G)$ of a multigraph $G = (V, E)$ has ground set E and independent sets F for which (V, F) is a forest. Its loops are exactly those of G, and parallel edges remain distinct elements. An independence oracle decides whether a given subset of the ground set is independent. We use such oracles only for graphic matroids of quotients on four vertices.

The Characterization. The following theorem characterizes the label-connected multigraphs by the two witness types. We prove sufficiency in Section 3 and necessity in Section 4.

▶ **Theorem 2.3** ([3, 4, 11]). *Let G be a multigraph. Then G is a yes-instance of* Gossiping *if and only if at least one of the following conditions holds:*

1. *G has at most one vertex;*
2. *G is a T-graph;*
3. *G is an M-graph.*

3 Sufficiency: Constructing a Temporal Labeling

We first describe how the components of the two forests meet the cycle of an M-witness in an M-graph. An *interval* of the cycle is a nonempty set of consecutive cycle vertices, forming a path; a single vertex is allowed. The following lemma states that the four components of each forest meet the cycle in the same four intervals. The four edges between consecutive intervals can be paired by taking every other edge in cyclic order. Each pair is a matching whose endpoints provide one root in each component of either forest. We prove this below.

▶ **Lemma 3.1** ([11, Corollary 1 and proof of Theorem 3]). *Let G be an M-graph, and let (C, F_1, F_2) be an M-witness of G. There is a partition of $V(C)$ into four nonempty consecutive intervals X_1, X_2, X_3, X_4, indexed in cyclic order, with the following properties:*

- Common intervals. *For each forest F_j, the intersections of its four components with $V(C)$ are exactly X_1, X_2, X_3, X_4.*
- Alternating matchings. *Let b_i be the cycle edge joining X_i to X_{i+1}, with indices modulo four. Then $B_1 = \{b_1, b_3\}$ and $B_2 = \{b_2, b_4\}$ are edge-disjoint matchings. Every component of each forest contains exactly one vertex of $V(B_1)$ and exactly one vertex of $V(B_2)$.*

Krumme calls $b_1, \ldots, b_4$ the *essential* edges. If an interval consists of a single vertex, then B_1 and B_2 share that vertex. Thus the two matchings need not have disjoint vertex sets. In particular, both forests of an M-witness of an M-graph can be rooted[2] at the same four vertices, namely at the endpoints of B_1. The labeling construction of Lemma 3.2 instead roots F_1 at the endpoints of B_1 and F_2 at those of B_2.

The classical constructions of Burosch and Laborde [3, Lemmas 7–8] and Krumme [11, Theorem 3] turn either witness into a call schedule. We use the following explicit running-time guarantee.

▶ **Lemma 3.2** (Linear-time implementation of the classical construction). *Given a T-witness or an M-witness of G, a temporal labeling under which G is temporally connected can be constructed in $\mathcal{O}(n + m)$ time.*

[2] Rooting a forest means choosing one vertex as the root of each tree component.

For each forest F_j of an M-witness with cycle C, contract each component of F_j in $C \cup F_j$, retaining all cycle edges, including any resulting loops. This produces a connected multigraph on four vertices. Choosing three cycle edges that form a spanning tree of this quotient completes F_j to a spanning tree of the original graph. The corresponding graphic matroids therefore have rank three. The next lemma helps us coordinate these choices for the two forests, including when the graph also admits a T-witness.

A *coloop* of a matroid is an element belonging to every basis. In a graphic matroid, coloops correspond to bridges: edges that every spanning tree must use. Every cut of either quotient corresponds to a cut of the cycle and therefore has even size. Hence neither quotient has a bridge. Thus no cycle edge is forced into every completion of either forest, which is why the next lemma assumes that neither matroid has a coloop.

▶ **Lemma 3.3.** *Let M_1, M_2 be rank-three matroids on the same ground set E, neither having a coloop. Given independence oracles for M_1 and M_2, in $\mathcal{O}(|E|)$ oracle calls one can either find bases J_i of M_i with $|J_1 \cap J_2| \leq 1$, or find a four-element set $W \subseteq E$ such that, in each matroid, the bases are exactly the three-element subsets of W.*

Proof. All bases below are found greedily. Whenever we construct a basis in one matroid, we compare it with every basis already constructed in the other matroid and return if some intersection has size at most one.

Fix a basis $B = \{y_1, y_2, y_3\}$ of M_2. Since M_1 has no coloop, for each i it has a basis A_i avoiding y_i. Unless we have already returned,

$$A_i = (B \setminus \{y_i\}) \cup \{x_i\} \qquad \text{for some } x_i \notin B.$$

For each h, choose a basis B_h of M_2 avoiding y_h. For distinct i, j, h, the basis B_h must contain $\{y_j, x_i\}$ to meet A_i twice, and $\{y_i, x_j\}$ to meet A_j twice. Since $|B_h| = 3$, this implies $x_i = x_j$. Thus all x_i are the same element x, and $B_h = (B \setminus \{y_h\}) \cup \{x\}$. Set $W := B \cup \{x\}$.

Next choose a basis A_0 of M_1 avoiding x. If no comparison returns a pair of bases, then A_0 must contain $B \setminus \{y_h\}$ for every h, and hence $A_0 = B$. Consequently, both matroids contain all four three-element subsets of W as bases.

It remains to inspect $E \setminus W$. If some $e \in E \setminus W$ is a non-loop of either matroid, extend e to a basis A of that matroid. Then $|A \cap W| \leq 2$. Choosing $w \in A \cap W$ when this intersection is nonempty, and any $w \in W$ otherwise, gives $|A \cap (W \setminus \{w\})| \leq 1$; the set $W \setminus \{w\}$ is a basis of the other matroid, so we return this pair. Otherwise, every element outside W is a loop in both matroids, and their bases are exactly the three-element subsets of W.

We perform only constantly many greedy basis computations and one scan of $E \setminus W$. Each greedy computation uses $\mathcal{O}(|E|)$ oracle calls, which proves the claimed bound. ◀

The rank-three lemma gives a linear-time procedure for finding either two almost edge-disjoint spanning trees or the common interval partition needed by the cycle schedule.

▶ **Lemma 3.4.** *Given an M-witness (C, F_1, F_2), in $\mathcal{O}(n)$ time one can either find two spanning trees sharing at most one edge, or find four edges of C giving the common interval partition of Lemma 3.1.*

Proof. For each j, identify vertices of C belonging to the same component of F_j, retaining the individual cycle edges and any resulting loops. The resulting quotient graph Q_j is connected on four vertices. Every cut has even size, since it comes from a cut of a cycle, so Q_j has no bridge. Its graphic matroid M_j has rank three and no coloops, regardless of whether G is a T-graph. A basis of M_j completes F_j to a spanning tree, and two such trees

intersect exactly in their chosen cycle edges. Apply Lemma 3.3 to M_1 and M_2. If it returns two bases whose intersection has size at most one, these bases complete F_1, F_2 to the required spanning trees. Otherwise it returns a four-element set W such that every edge outside W is a loop in both quotients.

Deleting the four edges of W from C leaves four nonempty paths, allowing single-vertex paths. Each lies in one component of each forest. Since all four components of each forest meet C, the paths belong to distinct components in both forests. Thus their vertex sets give the common four-interval partition.

The quotient has only four vertices, so each independence test takes constant time. Hence Lemma 3.3 takes $\mathcal{O}(|E(C)|)$ time. Computing the forest components takes $\mathcal{O}(n)$ time. This proves the bound. ◀

Proof of Lemma 3.1. Apply Lemma 3.4 to the given witness. Since G is an M-graph, it is not a T-graph. The procedure therefore cannot return two spanning trees sharing at most one edge and must return the four intervals of the first statement. Since the intervals are pairwise disjoint, b_1, b_3 are vertex-disjoint, and so are b_2, b_4. Each interval X_i contains exactly one endpoint of b_1 or b_3 and exactly one endpoint of b_2 or b_4. The second statement follows, since the components of each forest meet $V(C)$ exactly in $X_1, \ldots, X_4$. ◀

We now prove Lemma 3.2: given either type of witness, we construct a temporally connecting labeling in $\mathcal{O}(n + m)$ time. In the call interpretation, we first gather secrets at roots, exchange them between roots when necessary, and finally distribute them to all vertices.

Proof of Lemma 3.2. For two edge-disjoint spanning trees, choose an arbitrary common root. Gather all secrets along the first tree, calling each nonroot vertex's edge to its parent after all edges from that vertex to its children. Then distribute the secrets along the second tree, calling each nonroot vertex's edge to its parent before all edges from that vertex to its children. If the trees share one edge uv, delete that edge from both trees. Gather along the two components of the first tree toward u and v, make the call uv, and distribute along the two components of the second tree rooted at u and v. In either case every selected edge is used once.

Now suppose an M-witness is given. Apply Lemma 3.4. If it returns two spanning trees, use the preceding construction. Otherwise it returns the common four-interval partition directly, even if G happens to be a T-graph for another reason. Index the four intervals $X_1, \ldots, X_4$ in cyclic order, and let $b_1, \ldots, b_4$ be the essential edges as in Lemma 3.1, that is, b_i joins X_i to X_{i+1}, with indices modulo four. In each interval X_i, let p_i be the endpoint incident with $\{b_1, b_3\}$ and q_i the endpoint incident with $\{b_2, b_4\}$. They coincide if the interval has just one vertex. First gather along the blue forest F_1, rooting its component meeting X_i at p_i. Call b_1, b_3. Next, in each interval, call the edges of its path in order from p_i to q_i. At this point q_1, q_2 know the secrets from the first two components of F_1, and q_3, q_4 know those from the other two. Call b_2, b_4; all four vertices q_i now know every secret. Finally distribute along the orange forest F_2, rooting its component meeting X_i at q_i. Figure 2 shows the resulting schedule on the running example.

The selected edges are pairwise distinct. Label them by their positions in the gossip scheme, and label any remaining edges later in arbitrary order. Additional calls cannot destroy temporal connectivity. All forest traversals and cycle traversals take $\mathcal{O}(n)$ time. Together with Lemma 3.4 and the $\mathcal{O}(m)$ time to label unused edges, this gives the claimed bound. ◀

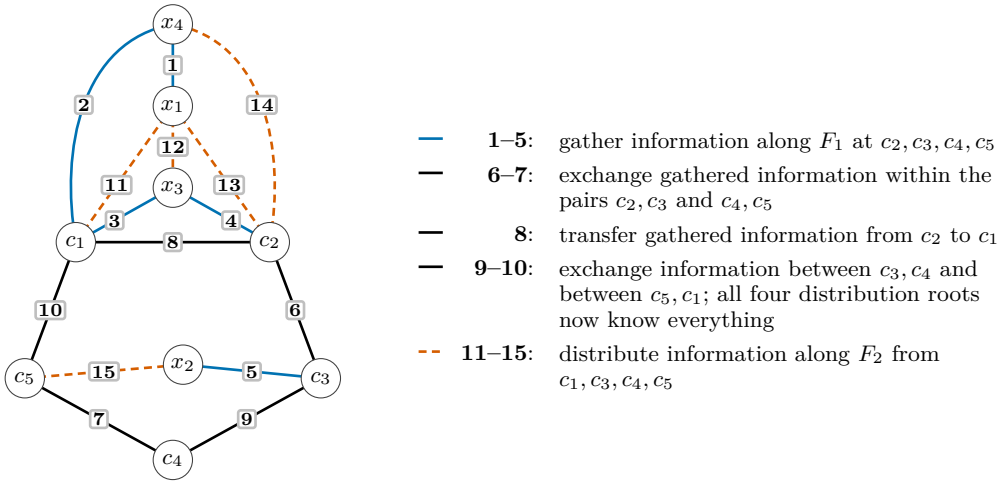


Figure 2 A temporally connecting labeling of the running example from Figure 1, obtained by the construction of Lemma 3.2. Each edge is labeled by its position in the gossip scheme. Blue solid edges gather information at c_2, c_3, c_4, c_5; black cycle edges exchange it so that the distribution roots c_1, c_3, c_4, c_5 know everything after call 10; orange dashed edges distribute it along F_2. Call 8 transfers information within the only interval with more than one vertex, from c_2 to c_1. Every edge is used exactly once.

In particular, Lemma 3.2 proves the sufficiency direction of Theorem 2.3. The algorithm accepts an arbitrary M-witness and does not require a separate test for whether the graph is a T-graph.

4 Necessity

We prove the necessity direction of Theorem 2.3. Graphs with at most one vertex satisfy the characterization directly. For a graph with at least two vertices, we construct a T-witness or an M-witness from a complete call order. The proof has four stages.

1. *Construct the quotient and bound its size.* Contract maximal sets supporting two edge-disjoint spanning trees. An exchange argument bounds the number of edges in the quotient and gives a T-witness when the upper bound is attained (Sections 4.1 and 4.2).
2. *Establish the lifting condition.* To lift a quotient M-witness, inside each block on its cycle we need two spanning trees and a path between its attachment vertices, all edge-disjoint. We establish a sufficient condition for finding these subgraphs (Section 4.3).
3. *Use the call order to obtain canonical structure.* The call order gives a lower edge bound. At equality, it yields a canonical division into early and future calls (Section 4.4).
4. *Select a compatible cycle and lift the witness.* Use the canonical structure to select a four-cycle satisfying the lifting condition, extend it to a quotient M-witness, and lift it to the original graph (Sections 4.5 and 4.6).

Figure 3 summarizes these steps and the case distinctions between them.

4.1 Two-Tree Blocks and Their Quotient

We first construct the quotient and show that its T-witnesses lift to the original graph. Fix a multigraph G with a complete call order. Within this argument graphs are loopless: deleting loops preserves completeness, since they transmit no new secret. Paths may be trivial, and

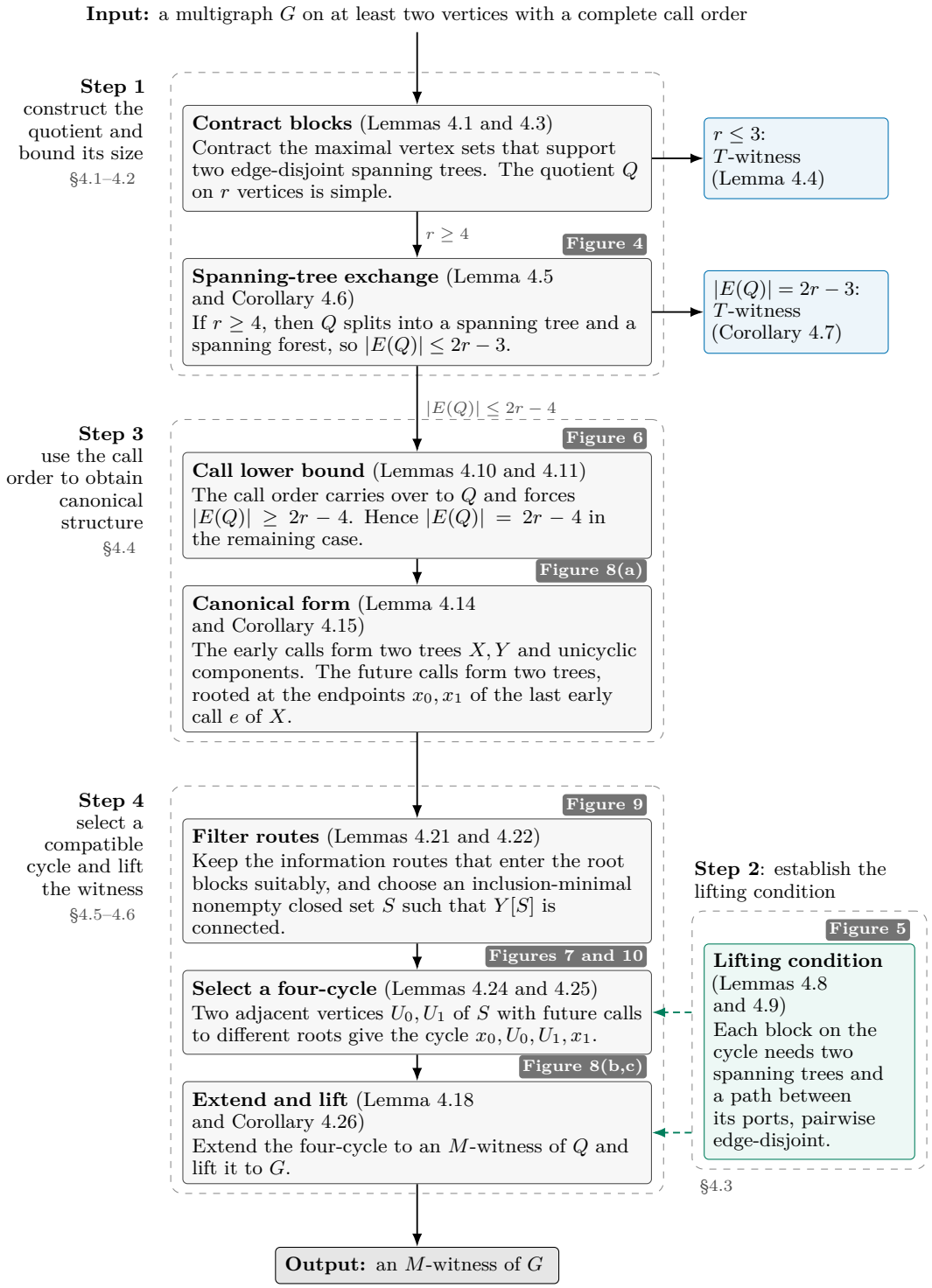


Figure 3 Roadmap of the necessity proof of Theorem 2.3. The dashed boxes are the four steps listed at the beginning of Section 4. Small quotients and the upper edge count give a T-witness (blue). Otherwise, the call order fixes the edge count and the canonical structure, which guides the choice of a four-cycle that satisfies the lifting condition of Step 2 (green, dashed arrows). The gray tags name the figures illustrating each part.

spanning subgraphs retain isolated vertices. For an edge set A, the notation $H[A]$ denotes the spanning subgraph with edge set A; for a vertex set it denotes the induced subgraph. Deleting a path means deleting its edges. A path is *nonseparating* if this deletion leaves its ambient graph connected.

In a graph H, a nonempty vertex set $X \subseteq V(H)$ is a *two-tree set* if $H[X]$ contains two edge-disjoint spanning trees. It is *nontrivial* if $|X| \geq 2$. A singleton is a two-tree set, with two empty trees. These are the vertex sets of the T^*-subgraphs of Burosch and Laborde [3].

▶ **Lemma 4.1.** *If X and Y are two-tree sets and $X \cap Y \neq \varnothing$, then $X \cup Y$ is a two-tree set.*

Proof. Choose edge-disjoint spanning trees A_X, B_X of $G[X]$ and A_Y, B_Y of $G[Y]$. The graph with edge set $E(A_X) \cup E(A_Y)$ is connected on $X \cup Y$. Extend A_X to a spanning tree A of this graph. Every edge of A outside A_X belongs to A_Y and has at least one endpoint outside X: otherwise it would form a cycle with the spanning tree A_X on X.

Consider the graph on $X \cup Y$ with edge set

$$E(B_X) \cup \big(E(B_Y) \setminus E(A_X)\big).$$

It is edge-disjoint from A. Indeed, B_X avoids A_X, and its edges have both endpoints in X, so it also avoids $A - A_X$. The retained edges of B_Y avoid A_X by definition and avoid $A - A_X$ because the latter edges belong to A_Y. The graph is connected: every deleted edge of B_Y belongs to A_X and has both endpoints in $X \cap Y$, so its endpoints can instead be joined through B_X. Thus it contains a spanning tree B edge-disjoint from A. ◀

We use this union property to decompose the graph into maximal two-tree sets.

▶ **Corollary 4.2.** *The inclusion-maximal two-tree sets form a partition $\mathcal{B}$ of $V(G)$.*

Proof. Every vertex belongs to a two-tree set and hence, by finiteness, to a maximal one. Two distinct maximal sets cannot intersect by Lemma 4.1. ◀

We call the sets in $\mathcal{B}$ *blocks*. Let Q be obtained by contracting each block to one vertex. Delete edges internal to a block, and retain each external edge as a distinct edge of Q, together with its original endpoints.

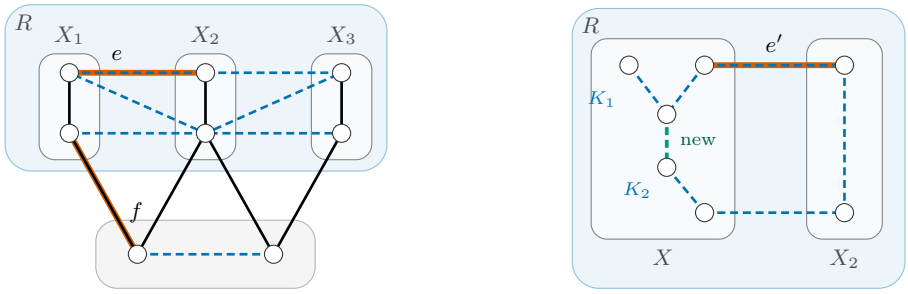
▶ **Lemma 4.3.** *The quotient Q is simple and contains no nontrivial two-tree set. If G is connected, then Q is connected.*

Proof. The quotient Q has no loops, since all edges internal to a block were deleted. Suppose a set of at least two vertices of Q supports two edge-disjoint spanning trees. Lift their edges to G. In each represented block, add one internal spanning tree to the first lift and the other to the second. Each lift is connected and has exactly one less edge than vertices, so the two lifts are edge-disjoint spanning trees on the union of the blocks. This contradicts their maximality. Two parallel edges of Q would themselves be two edge-disjoint spanning trees on their two endpoints, so no parallel edges occur. Connectivity is preserved by contracting connected sets. ◀

A T-witness of the quotient lifts to the original graph by using the two internal spanning trees in each block.

▶ **Lemma 4.4.** *If Q has a T-witness, then G has a T-witness. If Q has one vertex, then G has two edge-disjoint spanning trees.*

Proof. Lift each quotient tree and add one of the two internal spanning trees in every block. The two resulting graphs are spanning trees, and their intersection consists precisely of the lifts of shared quotient edges. The one-vertex case is the definition of a two-tree block. ◀



(a) a nonbridge e of $K[R]$ between pieces

(b) after replacing $S[X]$

Figure 4 The two steps in the proof of Lemma 4.5. Black edges belong to the spanning tree S, and blue dashed edges to its complement K. For a component R of K (blue region), the boxes are the vertex sets X_i of the components of $S[R]$. (a) If a complementary edge e joining two pieces were not a bridge of $K[R]$, the S-path between its endpoints would leave R through an edge f. Exchanging e and f keeps $K[R]-e$ connected and joins R to another component of the complement, reducing the number of components. (b) If $K[R]$ contains a cycle, it lies in one piece X, here in the component K_1 of $K[X]$. Replacing $S[X]$ by induction makes the complement inside X a forest that keeps every component of $K[X]$ connected and joins two of them. Here, the replacement removes a cycle edge inside K_1 and adds the green connection between K_1 and K_2. The old path between them leaves X through a bridge e' between pieces, which now lies on a cycle of the complement. The same argument is used in Lemma 4.18.

4.2 Spanning-Tree Exchange

The following lemma lets us replace a chosen spanning tree so that the remaining edges form a forest, without losing connections already present in the complement. Components of the complement may merge, but none split. This lets us bound the quotient's edge count and later modify trees while preserving specified complementary components. The lemma is adapted from Burosch and Laborde [3, Lemma 3]. A partition *coarsens* another if every part of the latter is contained in a part of the former.

▶ **Lemma 4.5.** *Let H be connected and loopless, with no nontrivial two-tree set. For every spanning tree T of H, there is a spanning tree S such that $H - E(S)$ is a forest and its component partition coarsens that of $H - E(T)$.*

We choose a spanning tree whose complement has as few components as possible, subject to the coarsening condition. If the complement contains a cycle, we modify the tree inside that component using induction. The modified tree admits an exchange that reduces the number of complementary components, contradicting minimality. Figure 4 illustrates both steps.

Proof of Lemma 4.5. Induct on $|V(H)|$. The assertion is immediate for one vertex. Among the spanning trees satisfying the required coarsening condition, choose S so that $H - E(S)$ has as few components as possible. Such a tree exists because T satisfies the condition. Write $K = H - E(S)$.

Fix a component of K with vertex set R, and let $X_1, \dots, X_s$ be the vertex sets of the components of $S[R]$. Every edge e of $K[R]$ joining different sets X_i is a bridge of $K[R]$. Otherwise, the unique S-path between its endpoints must leave R, since its endpoints are in different components of $S[R]$. Choose an edge f of this path with one endpoint in R and the other outside R. Replacing f by e in S gives another spanning tree. Deleting e from

$K[R]$ does not disconnect it, whereas adding f joins it to another component of K. The new complementary partition therefore coarsens that of K and has fewer components, a contradiction. This bridge property holds for every minimizing tree S.

Suppose now that $K[R]$ contains a cycle. Since H is loopless, $|R| \geq 2$. None of its edges is a bridge, so the cycle lies entirely inside one of the sets X_i; call that set X. If $s = 1$, then $S[R]$ and a spanning tree of $K[R]$ would be two edge-disjoint spanning trees on R, contrary to the hypothesis. Hence $s \geq 2$, and in particular $|X| < |V(H)|$.

Apply the inductive hypothesis to the connected induced graph $H[X]$ and its spanning tree $S[X]$. Obtain a spanning tree S_X whose complement is a forest and whose complementary partition coarsens that of $K[X]$. Replace $S[X]$ in S by S_X, obtaining a spanning tree $\widetilde{S}$ of H. Its complementary partition is exactly that of K. Indeed, each old component of $K[X]$ remains connected, all edges outside $H[X]$ are retained, and all newly added complementary edges have endpoints in R; thus no old component splits or merges with a different old component. Moreover, the component vertex sets of $\widetilde{S}[R]$ are still $X_1, \ldots, X_s$.

Let q be the number of components of $K[X]$. Since it contains a cycle, it has at least $|X| - q + 1$ edges. The new complement in $H[X]$ has the same number of edges, because both replaced trees have $|X| - 1$ edges. Being a forest, it therefore has at most $q - 1$ components. Consequently, it connects two distinct old components of $K[X]$.

Choose vertices in these two old components and a simple path between them in $K[R]$. This path must leave X. It thus contains an edge e joining different sets X_i, which is a bridge of $K[R]$ by the bridge property. Removing e separates the two selected vertices. Every component of $K[X]$ lies on one side of this cut, and its vertices remain connected after the replacement. The edges outside $H[X]$, other than e, are unchanged. Hence the old connections on each side of the cut remain, while the new connection inside X joins the two sides without using e. The edge e is consequently not a bridge of $H - E(\widetilde{S})$. But $\widetilde{S}$ is also a minimizing tree, and e still joins different components of $\widetilde{S}[R]$. This contradicts the bridge property. Thus K is a forest. ◀

The partition into a spanning tree and a complementary forest bounds the number of edges.

▶ **Corollary 4.6.** *If H is connected and loopless, has $r \geq 2$ vertices, and contains no nontrivial two-tree set, then $|E(H)| \leq 2r - 3$.*

Proof. By Lemma 4.5, the edges partition into a spanning tree and a spanning forest. The forest has at least two components, since otherwise H itself would be a two-tree set. The two parts have at most $r - 1$ and $r - 2$ edges, respectively. ◀

When the quotient attains this bound, the complementary forest has exactly two components, which yields a T-witness.

▶ **Corollary 4.7.** *If the connected quotient Q has $r \geq 2$ vertices and $2r - 3$ edges, then G has a T-witness.*

Proof. The partition supplied by Lemma 4.5 consists of a spanning tree A and a spanning forest B with exactly two components. Choose an edge e of A joining these components. The trees A and $B + e$ share exactly e. Apply Lemma 4.4. ◀

4.3 Lifting Witnesses

Two internal spanning trees suffice to lift a quotient T-witness. For a quotient cycle, those same two trees may leave no internal path between its attachment vertices. The next lemma

isolates the additional condition needed for lifting.

▶ **Lemma 4.8.** *Suppose Q has an M-witness (D, L_1, L_2). For each block X represented on D, let p_X, q_X be the original endpoints in X of its two incident cycle edges. Suppose $G[X]$ has two spanning trees A_X, B_X and a p_X-q_X path P_X whose three edge sets are pairwise disjoint. Then G has an M-witness.*

Proof. In blocks off D, choose any two edge-disjoint internal spanning trees and call them A_X, B_X as well. Lift L_1, L_2 and add all trees A_X, B_X, respectively. Replacing each quotient vertex by an internal tree preserves the component partition. A cycle in the lift could not lie inside one block, whose internal edges form a tree. Its external edges would therefore project to a nonempty closed trail in the quotient forest, which is impossible. Thus each lift is a forest with exactly four components. The two lifted forests are edge-disjoint: their external edges come from the edge-disjoint quotient forests L_1, L_2, and their internal edges come from the edge-disjoint trees A_X, B_X in each block. Replace each vertex X on D by P_X. Paths in different blocks are vertex-disjoint, and each block occurs only once on D, so this gives a cycle of length at least four. A path may be trivial if $p_X = q_X$. The lifted cycle avoids both forests by the local edge-disjointness assumptions and the edge-disjointness of the quotient witness. Every forest component still meets the lifted cycle. ◀

To obtain the local condition in Lemma 4.8, we will find a nonseparating path on one side of a call threshold and a spanning tree on the other side. The two threshold graphs have disjoint edges.

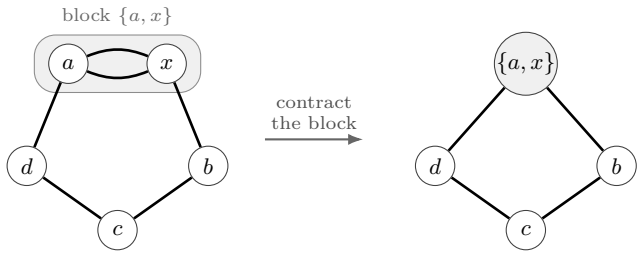
▶ **Lemma 4.9.** *Let J be a graph and let $E^-, E^+ \subseteq E(J)$ be disjoint. Suppose $(V(J), E^+)$ is connected and there is a p-q path P in $(V(J), E^-)$ such that $(V(J), E^- \setminus E(P))$ is connected. Then J contains two spanning trees and a p-q path whose edge sets are pairwise disjoint.*

Proof. Choose one spanning tree from $E^- \setminus E(P)$ and another from E^+. Both avoid P, and they avoid each other. ◀

Finding an M-witness in the quotient is not sufficient: its cycle may fail to lift while remaining edge-disjoint from the two forests. The following example explains why we must also control the cycle's attachment vertices inside each block. Consider the five-cycle (a, x, b, c, d, a) and add a second copy of the edge ax. The set $\{a, x\}$ is a two-tree block, and contracting it produces a four-cycle. The original graph has no T-witness: two spanning trees sharing at most one edge require at least seven edges, but it has six. It also has no M-witness: its only cycles of length at least four have length five, leaving just one edge for two forests that would each need one edge. Thus an M-witness of the quotient need not lift (Figure 5). The two internal edges of the block cannot supply both two spanning trees and a path from a to x.

4.4 Causal Orders and Canonical Form

We now use the complete call order to establish a lower bound on the quotient's number of edges and describe its structure when this bound is attained. Rather than fix one chronological order, we retain the comparisons between calls that preserve information propagation. This allows us to rearrange calls while keeping the scheme complete. We first define this partial order on the original calls and then restrict it to the quotient calls, so that the later lifting argument retains access to paths inside the original blocks.



(a) the graph G (b) the quotient Q

Figure 5 A quotient M-witness that does not lift. (a) The graph G has six edges on five vertices, so it has no T-witness. Its only cycles of length at least four have length five, leaving one edge for two forests, so it has no M-witness either. (b) The quotient Q is a four-cycle, which is an M-witness with empty forests. The two internal edges of the block cannot supply both two spanning trees and an a-x path. A third internal spanning tree, edge-disjoint from the first two, would supply such a path.

For a partial order, an *ideal* is a set closed under predecessors. A comparison $a < b$ is a *cover* if no element lies strictly between a, b. A *linear extension* is a total order respecting every comparison. We write $a < S$ when $a < s$ for every $s \in S$.

Fix a complete call order of G. Its *causal order* P is the transitive closure of the relations $a < b$ between chronologically ordered calls sharing an original endpoint. A linear extension of P preserves every information path and hence remains complete. Let P_Q be the restriction to the external calls, identified with $E(Q)$.

▶ **Lemma 4.10.** *Every linear extension of P_Q is complete on Q. Every cover of P_Q relates incident edges of Q.*

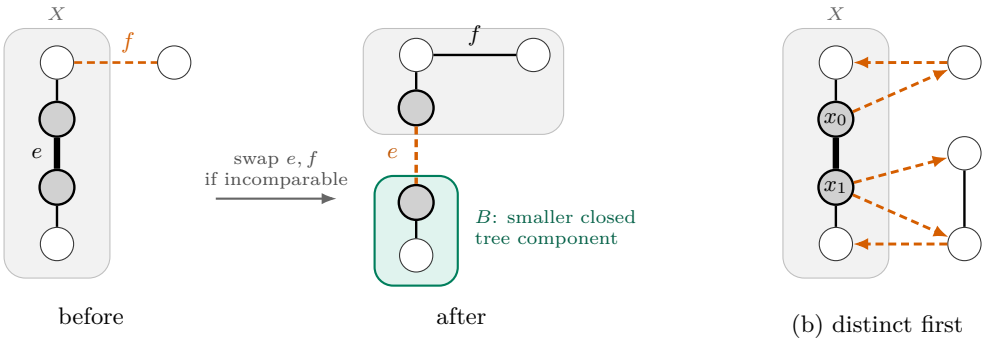
Proof. For any ordered pair of quotient vertices, choose an original vertex in each corresponding block and an increasing information path between them. Project this path to Q, omitting internal calls. The retained external calls increase in P_Q. Removing closed subwalks preserves this property. Thus P_Q is admissible: it provides an increasing path between every ordered pair of quotient vertices. Every linear extension preserves these paths.

For a cover $a <_{P_Q} b$, take an original causal chain from a to b. It has no intermediate external call, since that would lie strictly between a and b in P_Q. Its intermediate internal calls all belong to one block: successive calls sharing an original endpoint cannot move between blocks without an external call. Therefore a, b share that block in Q. The same conclusion holds for a chain with no intermediate call. ◀

This order must not be replaced by the usual causal order obtained by treating every quotient block as one vertex. Such a replacement would add comparisons between calls using different original attachment vertices, without providing internal paths between those vertices.

4.4.1 Closed-Tree Normalization and Counting

By Corollary 4.6, a quotient on $r \geq 4$ vertices has at most $2r - 3$ edges. We now show that completeness requires at least $2r - 4$ edges. The case of $2r - 3$ edges already yields a T-witness by Corollary 4.7, so only the case of $2r - 4$ edges remains for the subsequent M-witness construction. We rearrange the early calls to obtain a closed tree component. If this component is nontrivial, each of its maximal calls precedes every future call. Placing



(a) every maximal call of X precedes all future calls

(b) distinct first future calls

Figure 6 The closed-tree reduction (Lemma 4.11). Black edges are early calls in the ideal; orange dashed edges are future calls. (a) Let e be a maximal call of a minimum closed tree component X, and let f be a minimal future call. If e and f were incomparable, exchanging them would leave the part B of $X - e$ not touched by f as a smaller closed tree component. Hence e precedes every future call. (b) Placing the maximal call $e = x_0x_1$ last among the early calls, every other vertex still needs a future call. The first such calls (arrows, pointing to the vertex for which they are first) are pairwise distinct, which gives at least $r - 2$ future calls.

one such call last among the early calls lets us count distinct first future calls at all vertices except its two endpoints. A singleton component gives a separate, stronger bound. The closedness condition below enables the exchanges needed to obtain this division.

Let H be a finite simple graph on $r \geq 4$ vertices, with a partial order P whose covers relate incident edges and whose linear extensions are complete. The ideal-exchange argument follows Bumby [2] and Kleitman and Shearer [10]. Relative to an ideal J, we call its edges early calls and the remaining edges future calls. A linear extension placing J first realizes this division chronologically.

A tree component A of an ideal J is *closed* if there is no ideal $J' \supsetneq J$ such that $V(A)$ is the vertex set of a component of $H[J']$ and $(J' \setminus J) \cap E(H[V(A)]) \neq \varnothing$. Thus no larger ideal can add an internal edge while retaining this component's vertex set.

The following elementary observations will be used throughout.

1. A tree component is closed precisely when no minimal edge outside J is internal to it. Indeed, an internal minimal edge can be added; conversely, a saturated predecessor chain among the edges added by any larger ideal cannot move from an internal edge to an entirely outside edge, since covers are adjacent.
2. An edge maximal in a component of $H[J]$ is maximal in J, since a saturated chain in J cannot leave that component.
3. For a maximal edge $a \in J$ and a minimal edge $b \notin J$, a comparison $a < b$ is a cover; if they are incomparable, $(J \setminus \{a\}) \cup \{b\}$ is an ideal.
4. Closure persists under enlargement of the ideal while the component is retained.

If an ideal has two tree components, exchanges preserving its size produce an ideal with a closed tree component. To see this, take two tree components A, B. If neither is closed, choose a minimal remaining internal edge b of B and a maximal edge a of A. The two edges are incomparable: the reverse comparison contradicts ideality, whereas $a < b$ would be an adjacent cover between disjoint components. Swapping a, b splits A into two trees and makes B cyclic. Repeat with the two pieces of A. Their total vertex count is strictly smaller than $|V(A)| + |V(B)|$, so induction terminates. A singleton component is already closed.

▶ **Lemma 4.11** (Closed-tree reduction). *Let J be an ideal with a closed tree component A. There are an ideal I with $|I| \geq |J|$ and a closed tree component X with $V(X) \subseteq V(A)$ such that every minimal edge of $F = E(H) \setminus I$ crosses the boundary of X. If X is a singleton, then $|F| \geq r - 1$. Otherwise every maximal edge of X precedes every edge of F. Moreover, for any maximal edge $e = x_0x_1$ of X and any linear extension placing I first and e last in I, the first calls in F of the $r - 2$ vertices other than x_0, x_1 exist and are pairwise distinct. In particular, $|F| \geq r - 2$.*

The proof has three steps.

1. Choose a smallest closed tree component contained in $V(A)$ among ideals of size at least $|J|$, then enlarge the ideal while retaining that component. Every minimal remaining call must cross its boundary.
2. If the component is nontrivial, exchanging a maximal internal call with an incomparable minimal remaining call would produce a smaller closed tree component. Thus every maximal internal call precedes all remaining calls.
3. In the nontrivial case, place one such maximal call $e = x_0x_1$ last in the early ideal. Every vertex other than its endpoints still needs a future call. The order constraints prevent one future call from being the first for two such vertices, giving the bound $|F| \geq r - 2$.

Figure 6 illustrates the second and third steps.

If the component is a singleton, spreading its secret to all other vertices directly gives the stronger bound $|F| \geq r - 1$.

Proof of Lemma 4.11. Among ideals of size at least $|J|$ with a closed tree component contained in $V(A)$, choose one with a minimum-size such component X. This family is nonempty because it contains (J, A). Enlarge the chosen ideal maximally while retaining X and its edge set as a component; call the result I. Closure persists. A minimal edge outside I is not internal to X, by closedness, and is not entirely outside X, by maximality of I. Thus it crosses X.

If X is a singleton, its secret has reached nobody else at the end of I in an extension placing I first. Each future call informs at most one new vertex about that secret, so $|F| \geq r - 1$.

Suppose $|V(X)| \geq 2$ and let e be any maximal edge of X. For a minimal edge $f \in F$, the relation $f < e$ is impossible by ideality. If e, f were incomparable, interchange them. The edge f touches only one part of $X - e$; the untouched part B remains a tree component of $(I \setminus \{e\}) \cup \{f\}$. It is closed. Indeed, a minimal edge of the new complement is either e, an old minimal edge of the complement, or a cover-successor of f: for a newly minimal old complementary edge, all its former complementary predecessors must have been f. Any element strictly between f and that edge would also have been outside I and would still remain in the complement, a contradiction. The first crosses B, the second crosses X, and the third meets an endpoint of f, neither of which belongs to B. None is internal to B. This contradicts minimality of X, since $V(B) \subseteq V(A)$ and the ideal size is unchanged. Consequently $e < f$ for every minimal $f \in F$, and hence $e < F$.

Take any extension placing I first and $e = x_0x_1$ last in I. Every vertex other than x_0, x_1 needs a future call. A vertex outside X lacks the secrets of X; a vertex of $X - \{x_0, x_1\}$ lacks secrets from the opposite component of $X - e$, since only the endpoints of the last early call e receive those secrets at that call.

If a future call f were first at two vertices outside $\{x_0, x_1\}$, it would be minimal in F: otherwise a saturated chain from a predecessor in F to f would end with an earlier incident future call, contrary to first participation at both endpoints. But $e < f$ would then be a

cover, forcing f to touch an endpoint of e. This is a contradiction. The first future calls are therefore pairwise distinct. ◀

Starting with $r-2$ early calls, the reduction leaves at least $r-2$ future calls. Together, these give the classical lower bound.

▶ **Corollary 4.12** (Call lower bound). *The graph H has at least $2r-4$ edges.*

Proof. Completeness implies connectivity, so an ideal of size $r-2$ exists. Such an ideal has at least two tree components: if there are t tree components, its edge count is at least $r-t$. The preceding exchanges give a closed tree component without changing its size. Apply Lemma 4.11. The resulting ideal has at least $r-2$ edges and its complement has at least $r-2$ edges. ◀

If the total number of calls is exactly $2r-4$, an ideal containing at least $r-2$ calls and a closed tree component cannot be enlarged while retaining that component. Moreover, every minimal remaining call must cross its boundary.

▶ **Corollary 4.13** (Saturation of a closed early tree). *Suppose $|E(H)| = 2r-4$. If an ideal J has $|J| \geq r-2$ and a closed tree component A, then $|J| = r-2$, and every minimal edge outside J crosses the boundary of A.*

Proof. Lemma 4.11 gives $|J| \leq |I| \leq (2r-4)-(r-2) = r-2$, so equality holds. A minimal edge outside J cannot be internal to A, by closedness. If it were entirely outside A, adding it would retain A as a closed tree component in an ideal of size $r-1$, contradicting the just proved bound. Thus it crosses A. ◀

4.4.2 Canonical Form at Equality

It remains to handle a quotient on $r \geq 4$ vertices with exactly $2r-4$ edges. At this lower bound, every inequality in Lemma 4.11 is tight. We use this equality to divide the calls into early and future calls, with the future calls forming two trees ordered outward from their roots. This structure will guide the choice of a quotient cycle and the paths needed to lift it through the original blocks.

Figure 8 illustrates this canonical form and the subsequent construction and lifting of a quotient M-witness.

▶ **Lemma 4.14** (Canonical ideal normalization). *Let H be a finite simple graph on $r \geq 4$ vertices with $2r-4$ edges. Let P be a partial order on its edges such that every linear extension is a complete call order, and every cover relation of P relates incident edges. Then there is an ideal I with complement F, a tree component X of $H[I]$, and an edge $e = x_0x_1$ of X such that:*

1. *$|I| = |F| = r-2$ and $|V(X)| \geq 2$;*
2. *e is the unique maximal edge of X, and every other edge of X precedes e in P;*
3. *$e <_P f$ for every $f \in F$;*
4. *$H[F]$ is a spanning forest with two components rooted at x_0, x_1. Its edges are ordered outward: on each root-to-vertex path, successive edges increase in P;*
5. *every P-increasing trail ending at x_0 or x_1 contains at most one edge of $F \cup \{e\}$, and that edge is last.*

Moreover, every P-minimal edge of F crosses the boundary of X. Here an increasing trail has distinct edges, may repeat vertices, and its successive edges are strictly increasing in P.

Proof. An ideal of size $r-2$ has at least two tree components. The preceding exchanges produce an ideal J of that size with a closed tree component. Apply Lemma 4.11, obtaining I, X, F. The singleton case would give at least $2r-3$ edges, so X is nontrivial. The first-call injection and the total $2r-4$ force $|I| = |F| = r-2$. Fix a maximal edge $e = x_0x_1$ of X. In every linear extension placing I first and e last in I, each future edge is the first future call of exactly one nonroot. Choose one such extension.

Direct each future edge toward its uniquely assigned nonroot. In the chosen extension, its other endpoint is a root or has participated in an earlier future call. Starting with the two roots, the future calls therefore add one new vertex at a time. They form a spanning forest with exactly two rooted components, ordered outward in that extension. This outward order holds in P itself. Indeed, the first-call injection and equality argument applies to *every* linear extension putting I first and e last in I. In every such extension every prefix of the future calls is a forest each of whose nontrivial components contains a root. For the fixed graph $H[F]$, this forces every ancestor edge to occur before every descendant edge. All linear extensions of P restricted to F occur in such extensions of P: place I first in any extension ending at its maximal edge e. Therefore every ancestor edge precedes its descendants in P. Indeed, if such a pair were incomparable, adjoining its reverse comparison would remain acyclic and admit a linear extension, contrary to the order forced in every extension.

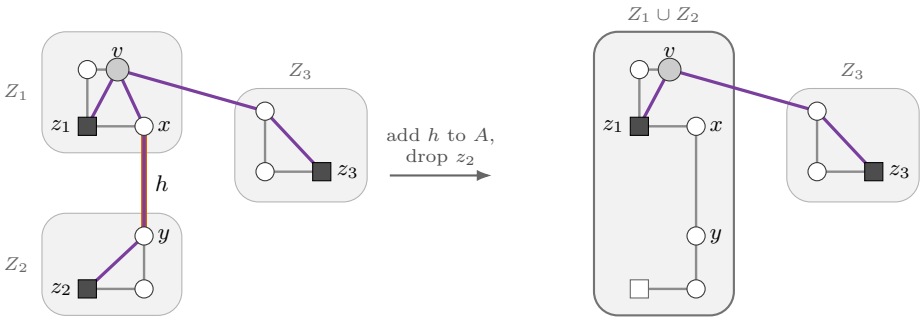
Both roots must participate in F, since X is a proper component of $H[I]$ and neither root yet knows secrets outside X. Let f_i be the first future edge at x_i. It is also first at its other endpoint, because every future edge is assigned to a nonroot and its root endpoint cannot be that assigned vertex. Hence f_i is minimal in F. It crosses the boundary of X. If s is any maximal edge of X, Lemma 4.11 gives $s <_P F$. Thus $s <_P f_0$ and $s <_P f_1$ are covers. Since s is internal to X, adjacency forces its endpoints to be x_0, x_1. Simplicity gives $s = e$. Finiteness then implies that every other edge of X precedes e.

Consider now an increasing trail ending at a root. Once it enters F, all its subsequent edges belong to F, because I is an ideal. A trail in a forest is a path. A path of at least two future edges ending at a root runs inward and therefore has decreasing consecutive edges, which is impossible. If the trail uses e, any subsequent early edge would have to be internal to X and follow e, also impossible. Its remaining suffix therefore lies in the tree $H[F] + e$. From an endpoint of e, future edges run outward, so a nonempty such suffix cannot end at either root. Consequently e, when used, is last and no future edge precedes it. This proves the final claim. ◀

▶ **Corollary 4.15** (The other early components are unicyclic)**.** *For the ideal I in Lemma 4.14, the graph $H[I]$ has exactly two tree components, one of which is X. All its other components are unicyclic. In particular, the second tree component Y is closed, and every minimal edge of F has one endpoint in X and the other in Y.*

Proof. There are at least two tree components because $|I| = r-2$. Every tree component other than X is closed: every minimal edge of F crosses the boundary of X and consequently cannot be internal to a different component. The closure criterion therefore applies. By Corollary 4.13, every minimal edge of F crosses the boundary of each such tree component as well as that of X. Three tree components would require a minimal edge of the nonempty set F to meet three disjoint vertex sets, which is impossible. There are exactly two tree components, and every minimal edge of F joins them.

For each connected component D of $H[I]$, set $\beta(D) = |E(D)| - |V(D)| + 1$. This is zero for a tree, one for a unicyclic component, and at least two for any other component. The



(a) the deepest crossing edge h (b) induction with $s-1$ components

Figure 7 The induction step of Lemma 4.16. Gray edges belong to A and form the components Z_i, each with a specified vertex z_i (squares); purple edges form a minimal tree T with $E(T) \subseteq D$ containing v and all z_i, rooted at v. (a) The crossing edge $h = xy$ of T with x deepest has its whole subtree below y inside one component, here Z_2, and only the v-z_2 path uses it. (b) Adding h to A merges Z_2 with the component of x, and z_2 is no longer specified. The path selected by induction avoids h and is also nonseparating in the original graph.

equality $|I| = r - 2$ gives

$$2 = \sum_{D} (1 - \beta(D)) = 2 - \sum_{D:\, \beta(D) \geq 1} (\beta(D) - 1).$$

Each summand on the right is nonnegative, so every nontree component has $\beta(D) = 1$, as claimed. ◀

4.5 Selecting Paths and Extending Cycles

To construct an M-witness that lifts to the original graph, we need two kinds of tools. Inside each block on the cycle, we must select a path between its attachment vertices while leaving enough edges for the two internal spanning trees. At the quotient level, we must extend the selected four-cycle to an M-witness without changing that cycle. We establish these tools here; Section 4.6 uses them to select a four-cycle with compatible attachment paths.

The first lemma selects a path whose deletion preserves connectivity. Its ordered version will let us apply this selection inside a block using calls before or after a specified call. Figure 7 illustrates the induction step in the proof of the static path-selection lemma.

▶ **Lemma 4.16.** *Let J be a graph with disjoint edge sets A, D. The graph $(V(J), A)$ has components $Z_1, \ldots, Z_s$, and each Z_i contains a specified vertex z_i. Suppose D connects a vertex v to every z_i. Then some v-z_i path in D can be deleted from J while leaving J connected.*

Proof. We induct on s. It suffices to retain A and a minimal tree T containing $v, z_1, \ldots, z_s$, with $E(T) \subseteq D$. If v is one of the specified vertices, choose the trivial path. If $s = 1$, deleting any v-z_1 path of T leaves the connected spanning graph $(V(J), A)$.

For $s \geq 2$, root T at v. Choose an edge $h = xy$ joining distinct components of $(V(J), A)$, with x the parent of y and x as deep as possible. Such an edge exists because T connects specified vertices in different components. By the choice of h, all vertices in the subtree rooted at y lie in the same component Z_i. Minimality of T ensures that this subtree contains a specified vertex, which must be z_i. Thus h is used by the v-z_i path and by none of the

other specified root paths. Let Z_j be the component containing x, where $j \neq i$. Add h to A, merging Z_i, Z_j. Discard z_i from the specified list and let D' be the union of the edge sets of the remaining root paths, keeping the full vertex set and $A \cup \{h\}$. The merged component retains z_j, and every other component retains its specified vertex. The remaining root paths avoid h, so induction applies with $s-1$ components and the disjoint connecting edge set D'. Its selected path also leaves the original graph J connected when deleted. ◀

In the ordered setting, gathering paths reach the endpoints of external calls before those calls occur, and connecting paths leave those endpoints afterward. Restricting to maximal external calls makes the two collections of internal edges disjoint. Paths between comparable external calls ensure that the retained endpoints still suffice for gathering, allowing us to apply the preceding lemma.

▶ **Lemma 4.17.** *Let J be a subgraph of a graph whose calls carry a partial order P. Let $t_1, \dots, t_k$ be external calls with endpoints z_i in J, and fix $v \in V(J)$. All comparisons and increasing paths below use the restriction of P to $E(J) \cup \{t_1, \dots, t_k\}$. Suppose:*

1. *every vertex of J has an internal increasing path to some z_i, all of whose calls precede t_i;*
2. *every z_i has an internal increasing path to v, all of whose calls follow t_i;*
3. *if $t_i < t_j$, there is an internal increasing path from z_i to z_j whose calls lie strictly between t_i and t_j.*

Then some z_i-v path can be deleted while retaining connectivity of J. The selected path uses only calls from the paths in (2).

Proof. Retain only maximal calls among $t_1, \dots, t_k$. Condition (3) extends every gathering path in (1) to the port of a maximal call, so these ports still suffice. Let A be the union of the edge sets of their gathering paths and D the union of the edge sets of their paths in (2). These edge sets are disjoint: a common call would satisfy $t_i < a < t_j$ for two maximal calls, a contradiction. Every component of $(V(J), A)$ contains a port. Choose one port in each component and apply Lemma 4.16. ◀

The next lemma completes the quotient forest construction once a compatible four-cycle has been selected. It does not select the original attachment paths.

▶ **Lemma 4.18.** *Suppose Q contains no nontrivial two-tree set, has $2r-4$ edges, and has the canonical partition I, F of Lemma 4.14. Let Y be its other early tree. Suppose $g = u_0u_1$ is an edge of Y and $f_i = x_iu_i \in F$ for $i = 0, 1$. Then the four-cycle with edges e, g, f_0, f_1 extends to an M-witness of Q whose two forests each have exactly one of x_0, x_1, u_0, u_1 in every component.*

Proof. Start with the spanning tree $B = Q[F] + e$. Its complement $K = Q[I \setminus \{e\}]$ has three distinguished tree components: the two parts A_0, A_1 of $X - e$ and the tree Y. Every other component is unicyclic by Corollary 4.15. Among spanning trees B containing e, f_0, f_1 but not g, whose complements have three distinct tree components containing x_0, x_1, u_0 and all other components unicyclic, minimize the number of complementary components. The initial tree $Q[F] + e$ satisfies these conditions. Write $K = Q - E(B)$ for the complement of the chosen tree. Since $g = u_0u_1$ remains in the complement, u_1 belongs to the component of u_0.

For a remaining unicyclic component Z, every edge of $K[V(Z)]$ joining different components of $B[V(Z)]$ is a bridge of $K[V(Z)]$. Otherwise add such a nonbridge edge h to B and remove a tree edge t on its fundamental cycle crossing the boundary of Z. The edge t is none of e, f_0, f_1: all their endpoints lie in the three distinguished components, whereas t crosses the boundary of Z. The edge g remains in the complement. The new complement

consists of $Z - h$ joined to another component by t. If that component is a tree, the merge is a tree; if it is unicyclic, the merge is unicyclic. None of the distinguished components merge with each other. Thus the three specified vertices remain in distinct tree components, and all other components remain unicyclic. This reduces the component count, a contradiction.

Since Z is unicyclic, the bridge property puts its unique cycle inside the vertex set W of one component of $B[V(Z)]$. The set W is a proper subset of $V(Z)$: otherwise $B[V(Z)]$ and a spanning tree of $K[V(Z)]$ would be two edge-disjoint spanning trees on $V(Z)$. Replace the tree $B[W]$ using Lemma 4.5 on $Q[W]$. The new internal complement is a forest whose partition coarsens the old one. If the old internal complement has q components, it has $|W| - q + 1$ edges, since exactly one component contains the unique cycle of Z. The new complement has the same number of edges and is a forest, so it has $q - 1$ components. Since its partition coarsens the old one, two old internal complementary components merge.

Globally the complementary vertex components remain unchanged, and the component Z stays connected. Its number of edges is unchanged, so it remains unicyclic. The vertex components of $B[V(Z)]$ also remain unchanged. Since $W \subseteq V(Z)$, the edges e, f_0, f_1, g are untouched, and the new tree satisfies the same minimizing conditions. But an old complementary path between the newly merged pieces leaves W and crosses an unchanged bridge joining distinct components of $B[V(Z)]$. The new internal connections preserve every old internal complementary component without using the bridge, since the bridge leaves W. Thus the old connections on either side of its cut remain, and the newly joined pieces connect the sides without the bridge. This makes that bridge a nonbridge, contradicting the bridge property for the equally minimizing new tree. Thus no unicyclic component remains.

The resulting complement $K' = Q - E(B)$ is a forest with $r-3$ edges and three components containing x_0, x_1, u_0 separately. Set

$$L_1 = K' - g, \qquad L_2 = B - \{e, f_0, f_1\}.$$

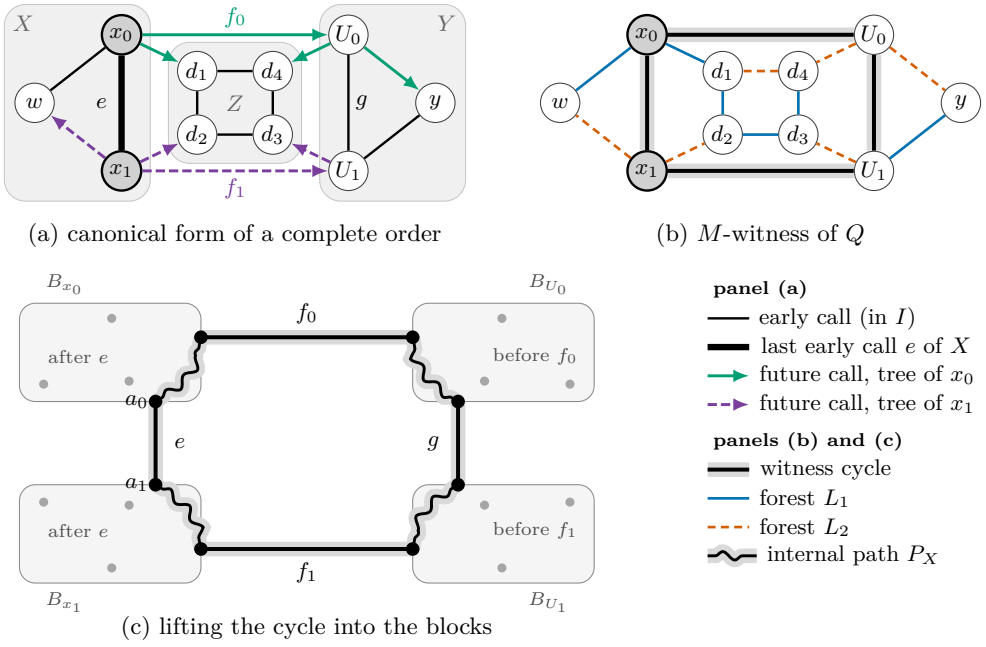
Since g lies in the tree component containing u_0, u_1, deleting it separates these two vertices. The first forest therefore has four components separating the four specified vertices. The second does too, because its three removed edges form the path u_0, x_0, x_1, u_1 in a spanning tree. Both avoid the four-cycle, and they are edge-disjoint from each other. Each of their components contains exactly one cycle vertex. ◀

4.6 Selecting Compatible Ports Using Original Routes

We now carry out the final stage of the necessity proof: select a four-cycle in the quotient whose witness can be lifted to the original graph. Finding the cycle alone is not enough. In each original block on the cycle, its two attachment vertices—called *ports*—must be joined by a path such that this path and two spanning trees of the block are pairwise edge-disjoint. We use information routes in the original graph to select the quotient edges and guarantee these internal paths. We then extend the four-cycle to an M-witness using Lemma 4.18 and lift that witness using Lemma 4.8.

We keep the original causal order P and its restriction P_Q to external calls. Throughout this section, Q has $r \geq 4$ vertices and $2r - 4$ edges, and $I, F, X, Y, e = x_0x_1$ are as in Lemma 4.14 and Corollary 4.15. Let B_U denote the original block represented by a quotient vertex U. For an external call a incident with U, write $p_U(a)$ for its original endpoint in B_U. In particular, let $a_i = p_{x_i}(e)$ for $i \in \{0, 1\}$.

An *original route* is a path in G whose successive calls increase in P. Completeness supplies such a route between every ordered pair of original vertices. Its external projection



(a) canonical form of a complete order

(b) M-witness of Q

(c) lifting the cycle into the blocks

Figure 8 The necessity proof of Theorem 2.3 on a quotient Q with $r = 10$ vertices and $2r-4 = 16$ edges that contains no nontrivial two-tree set. A complete call order is d_1d_2, d_3d_4, d_2d_3, d_4d_1, wx_0, yU_1, U_0U_1, x_0x_1, followed by x_0U_0, x_1U_1, x_0d_1, x_1d_2, U_0d_4, U_1d_3, x_1w, U_0y; every linear extension of its causal order is complete. (a) The canonical form of Lemma 4.14 and Corollary 4.15. The early calls form the tree components X and Y and the unicyclic component Z; the call $e = x_0x_1$ is the unique maximal call of X and precedes every future call. The future calls form two trees rooted at x_0 and x_1 and are ordered outward from the roots; the minimal future calls f_0, f_1 join X to Y. Here $f_i = f_{U_i} = x_iU_i$ for $i \in \{0,1\}$. (b) The M-witness of Q obtained from the four-cycle (x_0, U_0, U_1, x_1) by Lemma 4.18: the exchange moves the early call d_4d_1 into the spanning tree $Q[F] + e$ and the future call x_0d_1 out of it, which removes the unicyclic component. (c) Lifting to G (Lemma 4.8), shown schematically. Each quotient vertex represents an original block. To lift the witness, each block on the cycle needs two spanning trees and a path between its two ports—the original endpoints of its incident cycle edges—with all three edge sets pairwise disjoint. In B_{x_i}, the path uses calls after e, and deleting it leaves those calls connected, providing one spanning tree; the calls before e provide the other. In B_{U_i}, the path and the first spanning tree use calls before f_i, while the second spanning tree uses calls after f_i (Lemmas 4.9 and 4.25).

is a P_Q-increasing trail: its edges are distinct, although its quotient vertices may repeat. For an original block B and a call t, define

$$J_B^-(t) = \big(B, \{a \in E(G[B]) : a <_P t\}\big), \qquad J_B^+(t) = \big(B, \{a \in E(G[B]) : t <_P a\}\big).$$

These graphs retain every vertex of B. In the following, calls before or after t mean predecessors or successors of t in the causal order P. All paths below may be trivial. Deleting the edges of a trivial path leaves the graph unchanged.

The selection has three steps. First, retain routes whose final attachment to B_{x_i} admits an internal path whose deletion preserves the required connectivity. Second, use these routes to find a connected set of blocks in which internal return paths can be forced. Third, select two adjacent blocks with routes to different roots and compatible ports at both ends. We begin with the confinement and connectivity facts needed in the first two steps.

▶ **Lemma 4.19** (Confinement to a quotient forest). *Let $a <_{P_Q} b$ be external calls sharing a quotient vertex U. Suppose a quotient forest R contains a, b and every external call c with $a <_{P_Q} c <_{P_Q} b$. Then there is an internal original route from $p_U(a)$ to $p_U(b)$ whose calls lie strictly between a and b in P.*

Proof. Place a marker at both endpoints immediately after call a. A causal chain from a to b implies that at least one endpoint of b receives the marker before b. Use only the intermediate calls of this chain to transmit the marker. Tracing these transmissions gives an increasing original path, using calls strictly between a, b in P, from an endpoint of a to an endpoint of b.

Every external call of this path belongs to $R - \{a, b\}$. Deleting a, b from the forest separates their two nonshared endpoints from each other and from their common endpoint U. Thus the only possible match between an endpoint component of a and an endpoint component of b is the component containing U. The path therefore starts at $p_U(a)$ and ends at $p_U(b)$. Its external projection is a closed trail from U to U in a forest, so it is empty. The path is internal to B_U. ◀

We will apply this lemma in three settings. Comparisons between early calls in the same early tree are confined to that tree. Indeed, I is an ideal, and a causal chain between its external calls cannot leave an early component: consecutive external calls of a causal chain share a quotient vertex. Comparisons between future calls are confined to F, since I is an ideal. Finally, comparisons from e to future calls are confined to the tree $Q[F] + e$. To see this, e is maximal in I: a chain of incident early calls cannot leave X, and e is maximal in X. Hence no other early call lies above e.

▶ **Lemma 4.20** (Internal connectivity at the relevant thresholds). *For each $i \in \{0, 1\}$, the graphs $J^-_{B_{x_i}}(e)$ and $J^+_{B_{x_i}}(e)$ are connected. Moreover, if $U \in V(Y)$ and $f = Ux_i \in F$, then $J^+_{B_U}(f)$ is connected.*

Proof. Fix an original vertex $z \in B_{x_i}$ and an original route from z to a_{1-i}. Its first external call lies in I. Otherwise all its external calls would lie in F, and a trail in F could not connect its different root components. Denote this first call by a. The call a belongs to X and is incident with x_i. If $a = e$, the initial internal segment gathers z to a_i before e. Otherwise $a <_{P_Q} e$, and Lemma 4.19, applied in X, gives an internal route from $p_{x_i}(a)$ to a_i between a, e. Concatenate it with the initial internal segment. Erasing closed subwalks gives an internal route from z to a_i before e. These routes prove that $J^-_{B_{x_i}}(e)$ is connected.

Next take an original route from a_{1-i} to z. Lemma 4.14 applies to its external trail ending at x_i. Its last external call is either e or a future call t incident with x_i. Indeed, if there were no future call, the unique early-tree path from x_{1-i} to x_i would use e. If the last call is e, its internal suffix starts at a_i and lies after e. Otherwise $e <_{P_Q} t$, and confinement in $Q[F] + e$ gives an internal route from a_i to $p_{x_i}(t)$ between e, t. Append the internal suffix following t. This proves that $J^+_{B_{x_i}}(e)$ is connected.

Finally, fix $U \in V(Y)$ with $f = Ux_i \in F$ and an original vertex $z \in B_U$. Take an original route from a_i to z. Some external call of this route lies in F, since its endpoints belong to different early components. All external calls after the first future call remain in F. Let t be its last external call, which is incident with U. If $t = f$, its internal suffix gathers $p_U(f)$ to z after f. Otherwise t is an edge from U to one of its future children. The outward order gives $f <_{P_Q} t$. Confinement in F gives an internal route from $p_U(f)$ to $p_U(t)$ between f, t; append the suffix following t. Thus every original vertex of B_U is internally reachable from $p_U(f)$ after f. ◀

4.6.1 Filtering Routes

We first ensure that each root block B_{x_i}, for $i \in \{0,1\}$, contains two spanning trees and a path between its two cycle ports, with all three edge sets pairwise disjoint, as required by Lemma 4.8. A route entering B_{x_i} through a future call t determines the port $p_{x_i}(t)$. We retain only routes for which this port can be joined to the port a_i of e by a path in $J^+_{B_{x_i}}(e)$ whose deletion leaves that graph connected. The remaining edges of this graph supply one spanning tree, and $J^-_{B_{x_i}}(e)$ supplies the other. We show that every original source in a block represented in Y still has a retained route to each root block.

For $i \in \{0,1\}$, call a future edge $t = Vx_i$ with $V \in V(Y)$ *good at* x_i if there is an a_i-$p_{x_i}(t)$ path in $J^+_{B_{x_i}}(e)$ whose deletion leaves $J^+_{B_{x_i}}(e)$ connected. This property concerns only the edge t and its endpoint $p_{x_i}(t)$ in B_{x_i}.

▶ **Lemma 4.21.** *For every original vertex u in a block represented in Y and each $i \in \{0,1\}$, there is an original route from u to some original vertex of B_{x_i} whose last external call is good at x_i.*

Proof. For every original vertex $v \in B_{x_i}$, choose an original u-v route. Its external projection ends at x_i. The last assertion of Lemma 4.14 implies that this projection is an early path in Y followed by a single future edge $t_v = V_v x_i$. In particular, it cannot use e: an early prefix starting in Y cannot reach X.

The internal suffix from $p_{x_i}(t_v)$ to v lies after t_v. Confinement in $Q[F]+e$ gives an internal route from a_i to $p_{x_i}(t_v)$ between e, t_v. Whenever two selected future calls are comparable, confinement in F gives an internal route between their ports strictly between those calls.

Apply Lemma 4.17 with the call order reversed to the internal graph $J^+_{B_{x_i}}(e)$, with distinguished vertex a_i and external calls t_v. The reversed suffixes supply its gathering paths, the reversed routes from a_i to $p_{x_i}(t_v)$ supply its connecting paths, and the confined between-call routes supply its third condition. Repeated calls t_v are identified; all occurrences have the same endpoint in B_{x_i}, and their suffixes still cover all targets. Every used internal call lies after e. The lemma selects one t_v good at x_i, and the associated chosen original route proves the assertion. Its target may be any vertex of B_{x_i}; no claim that it ends specifically at a_i is needed. ◀

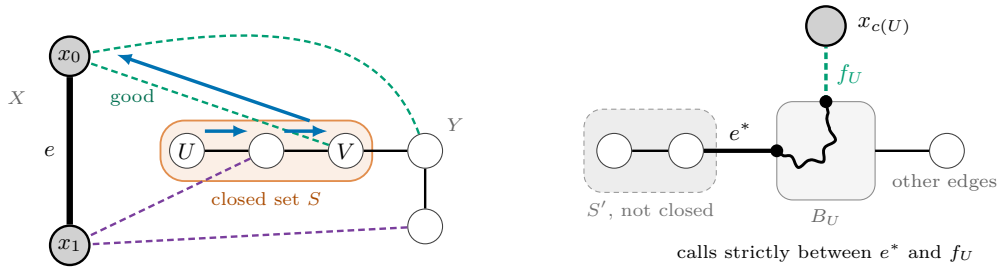
A *filtered route with root index i* starts at an original vertex of a block in Y, ends at any original vertex of B_{x_i}, and has a last external call good at x_i. Every such route projects to an early path in Y followed by that last future call. For $U \in V(Y)$, let $N_i(U)$ be the set of all vertices V that occur immediately before x_i in these projections, ranging over *all* filtered routes starting at *every* original vertex of B_U. A nonempty set $S \subseteq V(Y)$ is *closed* if

$$N_0(U) \cup N_1(U) \subseteq S \qquad \text{for every } U \in S.$$

Thus every filtered route starting in a block represented in S makes its final departure toward a root from a block still represented in S. The whole set $V(Y)$ is closed and induces a connected graph. Choose S inclusion-minimal among nonempty closed sets for which $Y[S]$ is connected. Figure 9 illustrates filtered routes and closed sets, together with the internal return paths obtained from the minimality of S.

4.6.2 Obtaining Internal Return Paths

Filtering provides suitable attachment paths inside the root blocks. We now prepare the corresponding paths inside blocks represented by the other early tree Y. Recall that S is



(a) a filtered route from U: $V \in N_0(U)$ (b) an internal return path in B_U

Figure 9 Filtered routes and minimal closed sets (Lemmas 4.21 and 4.22). (a) A filtered route projects to an early path in Y followed by one future call to a root block, here from V to x_0; its last call is good at that root. The vertex V is then in $N_0(U)$. A set S of vertices of Y is closed if every filtered route starting in a block represented in S makes its final departure toward a root from a block represented in S. (b) Let S be inclusion-minimal among nonempty closed sets inducing a connected subgraph of Y, and let $U \in S$. Let e^* be a maximal edge of $Y[S]$ at U. The vertex set S' of the component of $Y[S] - e^*$ not containing U is not closed, so some filtered route starting in S' leaves it. That route crosses e^* into the block B_U and leaves it through the future call f_U to a root. Its segment inside B_U (wavy) uses only calls between e^* and f_U.

an inclusion-minimal nonempty closed set for which $Y[S]$ is connected. We show that each block represented in S has a future call to a root and that every incident edge of $Y[S]$ can be followed by an internal route to the port of this future call. These routes will supply the connecting paths needed by the ordered path-selection lemma, Lemma 4.17.

▶ **Lemma 4.22** (Minimal closed sets supply internal returns). *The graph $Y[S]$ is a tree with at least two vertices. Every $U \in S$ has a unique future edge $f_U = Ux_{c(U)}$ incident with a root, for a root index $c(U) \in \{0,1\}$, and this edge is good at $x_{c(U)}$. For every edge h of $Y[S]$ incident with U, we have $h <_{P_Q} f_U$, and there is an internal original route from $p_U(h)$ to $p_U(f_U)$ with all calls strictly between h and f_U.*

Proof. By its choice, $Y[S]$ is a connected subgraph of the tree Y, hence a tree. A singleton closed set would require future edges from its sole vertex to both roots, since every original source has a filtered route of each root index by Lemma 4.21. Those edges would connect the two root components of F, which is impossible. Consequently $|S| \geq 2$.

Fix $U \in S$ and an edge e^* maximal among the edges of $Y[S]$ incident with U. Let S' be the vertex set of the component of $Y[S] - e^*$ not containing U. Then S' is a nonempty proper subset of S, and $Y[S']$ is connected. By minimality of S, the set S' is not closed. A witnessing filtered route has its starting block in S' and its exit in $S \setminus S'$. It crosses e^* into U and cannot use another early edge, by maximality of e^*. Its next external call is therefore a future edge f_U from U to a root. This call is good at that root, and the actual route gives $e^* <_{P_Q} f_U$, even if its internal segment is trivial. That segment is the required return path. There cannot be future edges from U to both roots because the roots lie in different components of F; simplicity gives uniqueness for the chosen root. This proves the assertion for every maximal e^*.

For an arbitrary incident edge h, choose a maximal incident edge e^* with $h \leq_{P_Q} e^*$. If $h \neq e^*$, confinement in the early tree Y supplies an internal original route from $p_U(h)$ to $p_U(e^*)$ between h, e^*. Append the return path just obtained between e^*, f_U. In particular $h <_{P_Q} f_U$. The ordered call intervals give an increasing internal walk; erase closed subwalks. The resulting path lies strictly between h, f_U. ◀

4.6.3 Selecting Neighbors with Different Root Indices

Each vertex $U \in S$ now has a future edge f_U to the root $x_{c(U)}$. We seek adjacent vertices U_0, U_1 with different root indices, so that their future edges, their common early edge, and $e = x_0x_1$ form a four-cycle. At both ends of the common early edge, we also need an internal path to the future port whose deletion leaves the corresponding graph $J^-_{B_U}(f_U)$ connected. We first find admissible neighbor choices at every vertex, then use a rerouting argument to obtain a pair that chooses each other and has different root indices.

▶ **Lemma 4.23.** *Fix $U \in S$. For every original vertex $u \in B_U$, choose a filtered route with root index $1 - c(U)$. Among their first external edges there is an edge h such that $J^-_{B_U}(f_U)$ contains a $p_U(h)$-$p_U(f_U)$ path whose deletion leaves this graph connected.*

Proof. The early projection of every chosen route stays in $Y[S]$ because its starting block and its exit belong to S. Its first external edge h_u is early: the only root exit from U has root index $c(U)$, whereas the chosen route has the other root index. Its initial internal segment gathers u to $p_U(h_u)$ before h_u. Lemma 4.22 supplies an internal route from $p_U(h_u)$ to $p_U(f_U)$ between h_u, f_U. All these gathering and connecting paths lie in $J^-_{B_U}(f_U)$.

For any comparable chosen first edges $h_u <_{P_Q} h_v$, confinement in Y supplies the internal between-port route required by the third condition of Lemma 4.17. Its first condition is supplied by the gathering prefixes, and its second by the return paths to $p_U(f_U)$. That lemma gives the claimed edge and nonseparating path. Repeated occurrences of the same first call may be identified; they have the same original port. ◀

To find an early edge with compatible internal paths at both ends, we encode the admissible choices at each block in a directed graph.

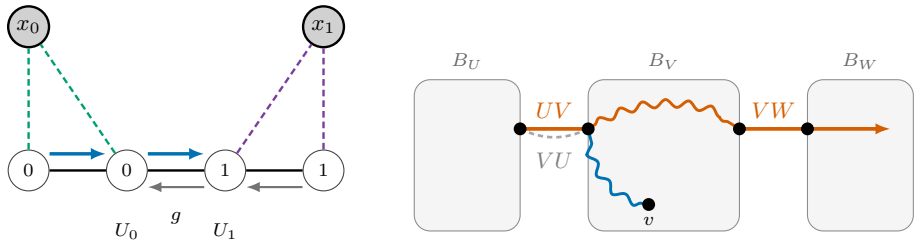
Define a directed graph D on S, using only edges of $Y[S]$. Include an arc $U \to V$ if there is a filtered route starting in B_U with root index $1 - c(U)$ whose first external edge is UV, and there is a $p_U(UV)$-$p_U(f_U)$ path in $J^-_{B_U}(f_U)$ whose deletion leaves that graph connected. The route and the nonseparating path need not use the same internal edges. Lemma 4.23 shows that every vertex of D has positive outdegree. Its formulation also allows us to prescribe a family of candidate routes before choosing the outgoing arc.

▶ **Lemma 4.24.** *If $U \to V$ is an arc of D and $c(U) = c(V)$, then V has an outgoing arc to a neighbor other than U.*

Proof. Take a filtered route witnessing $U \to V$. Its terminal exit has root index $1 - c(U)$, so its early projection cannot end at V. Let VW be its next external edge after UV. Between these calls the route has an actual internal segment in B_V from $p_V(UV)$ to $p_V(VW)$, with all calls strictly between UV, VW.

For every original vertex $v \in B_V$, choose a filtered route with root index $1 - c(V)$. If its first external edge differs from VU, keep it. Otherwise keep its internal prefix before VU, then append the actual internal V-segment of the U-route and the suffix of that route beginning with VW. The prefix ends at $p_V(VU)$ before VU; the replacement segment starts at that same port after UV. Hence this concatenation is an increasing original walk. Its first external call is VW, and its last external call is still good at the opposite root. The suffix after VW never returns to V, since its early projection is a path in Y. Erasing closed subwalks inside B_V therefore gives a filtered original route whose first external edge differs from VU.

We have obtained such a candidate route for every original vertex of B_V. Apply Lemma 4.23 to these candidates. The resulting outgoing arc of V avoids U. ◀



(a) the digraph D on the tree $Y[S]$ (b) rerouting when $c(U) = c(V)$

Figure 10 Selecting adjacent blocks with different root indices (Lemmas 4.24 and 4.25). (a) The vertices of the tree $Y[S]$ carry their root indices $c(U)$, and dashed edges show the future edges f_U to the roots x_0, x_1. Gray and blue arrows are arcs of D. A directed path that cannot be extended (blue) ends with an arc $U_0 \to U_1$; because $Y[S]$ is a tree, its last vertex can only point back to its predecessor, so $U_1 \to U_0$ is also an arc. If the two root indices were equal, rerouting would give another outgoing arc, so they differ, and $g = U_0U_1$ is the required edge. (b) The rerouting step: a route from $v \in B_V$ that would leave through VU retains its internal prefix (blue), then follows the internal segment and the suffix of a route witnessing $U \to V$ (orange), and leaves B_V through VW.

Figure 10 illustrates this rerouting step and the selection of adjacent blocks with different root indices in the next lemma.

▶ **Lemma 4.25** (Compatible four-cycle). *There is an edge $g = U_0U_1$ of Y with $f_{U_i} = U_ix_i$ for $i \in \{0, 1\}$ such that the four-cycle*

$$x_0, U_0, U_1, x_1, x_0$$

has the local lifting property of Lemma 4.8 in each of its four original blocks.

Proof. Choose a directed path of D that cannot be extended at its last vertex without repeating a vertex. Since every vertex has positive outdegree, this path has a last arc $U \to V$. Every outgoing neighbor of V lies on the path. A neighbor other than its predecessor U would form a cycle with a segment of that path in the tree $Y[S]$. Thus U is its only possible outgoing neighbor, and $V \to U$ is an arc. If $c(U) = c(V)$, Lemma 4.24 gives an outgoing neighbor of V other than U, a contradiction. Relabel the endpoints as U_0, U_1 so that $c(U_i) = i$, and let $g = U_0U_1$. Both orientations of g belong to D.

For each i, the arc $U_i \to U_{1-i}$ supplies a path between the U_i-ports of g and f_{U_i} whose deletion leaves $J^-_{B_{U_i}}(f_{U_i})$ connected. Lemma 4.20 gives a connected internal graph strictly after f_{U_i}. Apply Lemma 4.9 with these two disjoint edge sets. The block B_{U_i} therefore contains two spanning trees and a path between its cycle ports, with all three edge sets pairwise disjoint.

By Lemma 4.22, the edge f_{U_i} is good at x_i. Hence the ports of e and f_{U_i} in B_{x_i} are joined by a path whose deletion leaves $J^+_{B_{x_i}}(e)$ connected. The graph $J^-_{B_{x_i}}(e)$ is connected by Lemma 4.20. Another application of Lemma 4.9, with the two sides of e interchanged, gives the same local lifting property in B_{x_i}. The four quotient vertices are distinct because U_0, U_1 lie in the other early component Y. ◀

The selected four-cycle has compatible internal paths in all four blocks. We can therefore extend it to a quotient M-witness and lift that witness to G.

▶ **Corollary 4.26.** *Under the standing hypotheses, G has an M-witness.*

Proof. Apply Lemma 4.18 to the edge U_0U_1 and the two future edges from Lemma 4.25. It extends their four-cycle to a quotient M-witness. The local lifting property just proved satisfies all hypotheses of Lemma 4.8, which gives an M-witness of G. ◀

We now complete the necessity proof by distinguishing the small quotients and the two possible edge counts for larger quotients.

Proof of necessity in Theorem 2.3. Discard loops and assume that G has at least two vertices. Completeness implies that G is connected. Form its maximal two-tree blocks and their quotient Q. By Lemma 4.3, the graph Q is connected and simple and contains no nontrivial two-tree set. By Lemma 4.10, every linear extension of the induced order P_Q is complete on Q. If Q has one vertex, Lemma 4.4 gives two edge-disjoint spanning trees. If Q has two vertices, it is a single edge. If it has three, completeness forces a triangle: otherwise it is a two-edge path, and a secret sent on its second call cannot reach the endpoint whose only call was first. These quotients have two spanning trees sharing one edge, which lift by Lemma 4.4.

Suppose Q has $r \geq 4$ vertices. By Lemma 4.10 and Corollaries 4.6 and 4.12, we have $2r - 4 \leq |E(Q)| \leq 2r - 3$. The upper case gives a T-witness of G by Corollary 4.7. In the lower case, Lemma 4.14 and Corollary 4.15 give the standing hypotheses of this section. Corollary 4.26 then gives an M-witness of G. ◀

This completes the characterization. For sufficiency, either witness yields a complete call order by Lemma 3.2. For necessity, contracting the two-tree blocks reduces the problem to a simple quotient. Small quotients and the upper edge-count case yield a T-witness. In the remaining case, the call order supplies the canonical structure used to select a four-cycle with compatible internal paths. Extending this cycle to a quotient M-witness and lifting it through the blocks yields an M-witness of the original graph. Thus every graph admitting a temporally connecting labeling has one of the structures in Theorem 2.3.

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